ATTACK DEMONSTRATION
VULNERABLE SYSTEM

Sensor Spoofing Attack on Unprotected IoT Infrastructure

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VULNERABLE SYSTEM: Pi #7 — 192.168.50.100 (VLAN2)

⚡ ATTACKER — Suricata Laptop (192.168.50.191 VLAN2)

```
$ python3 attack_spoof.py --dual --pressure 0.0 --rate 0.1
```

Sends fake pressure readings: 0.0 PSI at 10 Hz (one request every 0.1 seconds)

```
HTTP POST /sensor → {"pressure": 0.0, "source": "attacker"}
```

Pi #7 — VULNERABLE BALLOON CONTROL SYSTEM | 192.168.50.100 | Raspberry Pi 5 | Flask API : 5000

Flask API /sensor

- ✗ NO AUTHENTICATION
- ✗ NO INPUT VALIDATION
- ✗ NO RATE LIMITING
- ✗ NO SENSOR CROSS-CHECK
- ✗ NO ANOMALY DETECTION

Accepts ALL data blindly

Control Loop

1. Receives: 0.0 PSI (FAKE!)
2. Target: 25 PSI
3. Deficit: 25 − 0.0 = 25 PSI
4. Decision: INFLATE! ⚠️

⚠️ Uses NETWORK DATA directly

⚠️ NO I2C sensor fallback

Relay Control (Sequent HATs)

8-Relay HAT (I2C 0x27):

Relay 2 — Inflate: OPEN

Relay 3 — Vent: CLOSED

Relay 4 — Relief: CLOSED

40A 3-Relay HAT (I2C 0x60):

Relay 3 — Pump: ON (23A)

PHYSICAL PRESSURE VESSEL (1L Bottle)

Real Pressure: 25 PSI → 26 PSI → 27 PSI → 28.7 PSI

System believes pressure is still 0.0 PSI — pump keeps running indefinitely

Pressure rises unchecked:

26 PSI ... 27 PSI ... 28 PSI ... 28.7 PSI

⚠ EXCEEDING SAFE LIMITS ⚠

⚠ CRITICAL FLAWS

- ✗ No authentication
- ✗ No input validation
- ✗ No sensor cross-check
- ✗ No anomaly detection

Result: 100% attack success

ATTACK PROGRESSION: SENSOR SPOOFING

T = 0 sec — NORMAL OPERATION

Real Pressure: 25.0 PSI

Network Reading: 25.0 PSI

Pump Status: OFF (at target)

System Belief: ✓ At target 25 PSI

Attack Traffic: None

T = 5 sec — ATTACK STARTS

Real Pressure: 25.2 PSI

Network Reading: 0.0 PSI (FAKE!)

Pump Status: ON (inflating)

System Belief: ✗ Too low! Need +25 PSI

Attack Traffic: 10 req/sec All accepted

T = 10 sec — PRESSURE EXCEEDS

Real Pressure: 26.5 PSI

Network Reading: 0.0 PSI (FAKE!)

Pump Status: ON (inflating)

System Belief: ✗ Too low! Need +25 PSI

Attack Traffic: 10 req/sec All accepted

T = 15 sec — FAILURE IMMINENT

Real Pressure: 28.7 PSI (!DANGEROUS!)

Network Reading: 0.0 PSI (FAKE!)

Pump Status: ON (inflating)

System Belief: ✗ Too low! Need +25 PSI

Attack Traffic: 10 req/sec All accepted

ATTACK SUCCESS METRICS

100%	10/sec	0	< 20s
success rate	**attack rate**	**rejected**	**to failure**
every request accepted	continuous spoofing	all accepted by system	total compromise

ATTACK OUTCOME: SYSTEM COMPROMISE

PHYSICAL CONSEQUENCES

BOTTLE STATUS (T = 15 sec)

Real Pressure: 28.7 PSI

System Belief: 0.0 PSI (spoofed)

Safe Limit: 25 PSI | Overpressure: +3.7 PSI

Burst Range: 39–71 PSI | ~30–60 sec to burst

FAILURE MODES

- ⚠️ Bottle rupture
- ⚠️ Valve damage | Sensor saturation
- ⚠️ No emergency shutoff

REAL-WORLD IMPACT

- Medical device attack
- Industrial control hack | Patient safety risk

DETECTION & LOGGING

SECURITY MONITORING

- ✗ NO intrusion alerts
- ✗ NO anomaly detection
- ✗ NO SIEM logging (Wazuh blind)
- ✗ NO authentication checks

WHY DETECTION FAILED

1. HTTP traffic appears legitimate
2. No baseline for normal behavior
3. No sensor cross-check

SURICATA IDS (post-hoc only)

✓ 150+ HTTP POST logged | Attacker: .191

⚠️ Too late to prevent — forensic value only

150+	0	< 20s	28.7
attacks sent	**detections**	**to failure**	**PSI reached**
all accepted	no alerts	total compromise	safe limit: 25 PSI